EDO STATE UNIVERSITY IYAMHO

Department of Computer Science

Faculty of Computing

FINAL YEAR PROJECT

CHAPTER-BY-CHAPTER WRITING GUIDE

Sections, Sub-sections, Standards & Requirements

Academic Session 2024/2025

Applicable to All 400-Level B.Sc. Computer Science, Software Engineering, and Cybersecurity Students

Version 1.0 | Issued by the Departmental Project Committee (DPC)

HOW TO USE THIS GUIDE

This guide specifies what must appear in each chapter of your project. Sections marked REQUIRED must be present in every project. Sections marked OPTIONAL should be included only if they are relevant to your specific topic — omitting them is not a weakness; forcing irrelevant sections in is.

Colour key used throughout this document:

- Green box = Required (all students)
- Teal box = Optional (include if applicable)
- Gold box = Important notices and warnings
- Blue box = Writing tips and guidance
- Red box = Common mistakes to avoid

CHAPTER ONE — INTRODUCTION

Chapter One sets the intellectual foundation of your entire project. It must convince the reader that your topic matters, that you understand the problem space, and that your project is both feasible and worthwhile. This chapter is NOT where you write your literature review or describe your methodology — those belong in Chapters Two and Three respectively.

⚠ IMPORTANT NOTICE

Chapter One must be written in clear academic English. Avoid informal language, first-person narrative (except in the Significance section where permitted), and vague generalities. Every claim made must either be supported by a citation or be demonstrably self-evident. A chapter that is merely descriptive and lacks critical engagement will receive a low score.

1.1 Overview of Sub-sections

Sub-Section	Applicable To	Status
1.1 Background to the Study	All topics	**Required**
1.2 Statement of the Problem	All topics	**Required**
1.3 Aim of the Study	All topics	**Required**
1.4 Objectives of the Study	All topics	**Required**
1.5 Research Questions	Research-based topics	**Required**
1.6 Research Hypotheses	Quantitative research	**Optional**
1.7 Significance of the Study	All topics	**Required**
1.8 Scope of the Study	All topics	**Required**
1.9 Limitations of the Study	All topics	**Required**
1.10 Definition of Terms	All topics	**Required**
1.11 Organisation of the Report	All topics	**Required**

1.1 Background to the Study

✓ REQUIRED — All Students

- Provide a broad, contextual narrative that introduces the problem domain.
- Move from the general (global context) to the specific (Nigerian/local context) to the particular (your exact problem area). This is known as the 'funnel structure'.
- Establish why the topic is relevant NOW — what recent trends, statistics, or events make it timely?
- Cite at least three (3) credible, recent sources (within the last 7 years) within this section.
- Typical length: 500 – 800 words.

✦ WRITING TIP

Think of the Background as a story that ends with the reader thinking: 'Yes, clearly something needs to be done about this.' Do not start with a dictionary definition. Start with a compelling fact, statistic, or real-world scenario relevant to your topic.

✗ COMMON MISTAKES TO AVOID

- Copying textbook definitions word-for-word as your opening paragraph.
- Writing a background that could apply to any computer science project — be specific.
- Ending the background without clearly signposting the problem your project addresses.

1.2 Statement of the Problem

✓ REQUIRED — All Students

- Clearly and precisely articulate the specific gap, deficiency, inefficiency, or challenge your project addresses.
- The problem statement must flow logically from the Background — the reader should feel the problem was revealed by the background, not dropped in suddenly.
- Use evidence (statistics, observed failures, user complaints, documented gaps in literature) to justify that the problem is real and significant.
- End with a concise one- or two-sentence 'problem statement sentence' that crystallises the exact problem.
- Typical length: 300 – 500 words.

✦ WRITING TIP

A strong problem statement answers three questions: (1) What exactly is wrong or missing? (2) Who is affected and how severely? (3) Why have existing solutions not adequately solved it? If you cannot answer all three clearly, refine the problem before proceeding.

✗ COMMON MISTAKES TO AVOID

- Stating a general area of interest ('The internet has many security problems') instead of a specific, localised problem.
- Writing a problem statement that does not connect to your proposed solution.
- Failing to cite evidence for the problem's existence.

1.3 Aim of the Study

✓ REQUIRED — All Students

- State a single, overarching aim — the broad goal your project seeks to achieve.
- Begin with an active infinitive verb: 'To develop…', 'To design…', 'To investigate…', 'To evaluate…'
- The aim should be achievable within one semester with the resources available to you.

- Typical length: 1 – 3 sentences.

Note: *The AIM is singular and high-level. The OBJECTIVES (Section 1.4) are the specific, measurable steps to achieve it. Do not confuse the two.*

1.4 Objectives of the Study

✓ REQUIRED — All Students

- List two to five (2–5) specific, measurable, achievable, relevant, and time-bound (SMART) objectives.
- Each objective must begin with an action verb: 'To identify…', 'To design…', 'To implement…', 'To test…', 'To evaluate…'
- The objectives must collectively cover the full scope of your project — from problem analysis through implementation and evaluation.
- Each objective should correspond to a visible deliverable or outcome in your project.
- At the end of Chapter Five, you will explicitly state whether each objective was achieved.

✦ WRITING TIP

A useful check: can you map each objective to a specific section in Chapters Three, Four, or Five? If an objective has no corresponding deliverable in the project, either remove it or create the deliverable.

1.5 Research Questions

✓ REQUIRED — All Students

- Formulate three to five (3–5) specific questions that your project seeks to answer.
- Research questions must be directly derivable from the objectives — each question should correspond to one or more objectives.
- Questions must be answerable through your methodology (system development, experimentation, survey, etc.).
- Frame questions precisely: avoid yes/no questions. Use 'What…?', 'How…?', 'To what extent…?', 'Which…?'

◈ OPTIONAL — Include Only if Applicable to Your Topic

- Projects that are purely systems-development without an empirical evaluation component may replace Research Questions with 'Design Questions' or 'Engineering Goals' — discuss with your supervisor before doing so.

1.6 Research Hypotheses

◈ OPTIONAL — Include Only if Applicable to Your Topic

- Include this section ONLY if your project involves formal hypothesis testing (e.g., comparing algorithm performance, testing user satisfaction statistically, or conducting controlled experiments).
- State the null hypothesis (H_0) and alternative hypothesis (H_1) for each testable claim.
- Hypotheses must be stated in a way that they can be accepted or rejected based on empirical data collected in Chapter Three and analysed in Chapter Four.
- If your project is a software development project without statistical experimentation, OMIT this section.

1.7 Significance of the Study

✓ REQUIRED — All Students

- Explain the practical, academic, and/or policy implications of your project.
- Address at least three distinct stakeholder groups who will benefit: e.g., end users, organisations, researchers, policymakers, the broader community.
- Be concrete — 'This system will reduce administrative errors in XYZ process by automating…' is stronger than 'This system will be useful to many people.'
- Typical length: 300 – 450 words.

1.8 Scope of the Study

✓ REQUIRED — All Students

- Define clearly what your project DOES cover — the boundaries of your system, dataset, geographic area, user group, or research context.
- State what is deliberately EXCLUDED and briefly explain why (resource constraints, time, ethical reasons, or deliberate focus).
- The scope must be realistic and consistent with what is actually delivered in Chapters Three and Four.
- Typical length: 200 – 350 words.

✗ COMMON MISTAKES TO AVOID

- Setting a scope so narrow that the project becomes trivial.
- Setting a scope so broad that it cannot be achieved in one semester.
- Listing limitations in the Scope section — limitations belong in Section 1.9.

1.9 Limitations of the Study

✓ REQUIRED — All Students

- Identify and honestly acknowledge the constraints that affected or may affect your project's outcome.

- Common categories: access to data, computational resources, sample size, geographic focus, time constraints, funding.
- For each limitation, briefly explain how you mitigated it or why it does not invalidate your findings.
- Typical length: 200 – 300 words.

Note: *Limitations are not failures — they demonstrate scholarly maturity. However, they must not be so severe that they undermine the entire project. If a limitation is that serious, it should have been addressed in the scope.*

1.10 Definition of Terms

✓ REQUIRED — All Students

- Define all technical, domain-specific, and context-sensitive terms used in your project that a non-specialist reader might not understand.
- Include acronyms and abbreviations used throughout the project (beyond standard CS terms).
- Definitions must be operational — i.e., reflect how the term is used in THIS project, not just a dictionary definition.
- Cite the source for each definition where appropriate.
- Typical number of terms: 8 – 20.

1.11 Organisation of the Report

✓ REQUIRED — All Students

- Provide a concise paragraph (or structured list) summarising what each chapter covers.
- This section serves as a roadmap for the reader.
- Typical length: 150 – 250 words.
- Example format: 'Chapter One presents the introduction... Chapter Two reviews related literature... Chapter Three describes the methodology...'

Chapter	Section	Min. Words	Max. Words
Chapter One	Background to the Study	500	800
Chapter One	Statement of the Problem	300	500
Chapter One	Aim & Objectives	150	300
Chapter One	Research Questions	150	300
Chapter One	Significance of the Study	300	450
Chapter One	Scope & Limitations	350	600
Chapter One	Definitions & Organisation	200	450

Chapter One	TOTAL (estimated)	1,950	3,400

CHAPTER TWO — LITERATURE REVIEW

Chapter Two demonstrates your scholarly engagement with the existing body of knowledge related to your topic. It is NOT a summary catalogue of what other people said. It is a critical, synthesised, thematic discussion that reveals what is known, what is debated, what is missing, and how your project fills a gap. A good literature review reads as a coherent academic argument, not as a series of disconnected summaries.

⚠ IMPORTANT NOTICE

The literature review must be current: at least 70% of your cited works must have been published within the last seven (7) years. Sources must be peer-reviewed journal articles, conference papers, or reputable books. Wikipedia, news blogs, and non-academic websites may not be used as primary sources. Minimum number of references in this chapter: 15.

2.1 Overview of Sub-sections

Sub-Section	Applicable To	Status
2.1 Introduction to the Chapter	All topics	**Required**
2.2 Conceptual Review	All topics	**Required**
2.3 Theoretical Framework	All topics	**Required**
2.4 Empirical Review of Related Works	All topics	**Required**
2.5 Review of Existing Systems/Tools	System-development topics	**Required**
2.6 Comparative Analysis of Related Works	All topics	**Required**
2.7 Identified Research Gap	All topics	**Required**
2.8 Summary of the Chapter	All topics	**Required**
2.9 Review of Relevant Algorithms	AI/ML/Data Science topics	**Optional**
2.10 Review of Relevant Standards/Protocols	Network/Security topics	**Optional**
2.11 Review of Relevant Datasets	Data-driven topics	**Optional**

2.1 Introduction to the Chapter

✓ REQUIRED — All Students

- Briefly state the purpose of the chapter and the themes it covers.
- Outline how the chapter is structured so the reader knows what to expect.
- Typical length: 100 – 200 words.

2.2 Conceptual Review

✓ REQUIRED — All Students

- Define and explain the core concepts and ideas underpinning your project.
- Go beyond dictionary definitions — explain how these concepts relate to one another and to your project.
- Use a conceptual framework diagram if it helps illustrate the relationship between key concepts.
- Draw from authoritative academic sources.
- Typical length: 600 – 1,000 words.

✦ WRITING TIP

For example, a project on 'Machine Learning for Malaria Diagnosis' would conceptually review: what machine learning is, how supervised classification works, what malaria is and its diagnosis challenges, and how image processing relates to medical diagnosis. Each concept should be explored with depth, not just defined.

2.3 Theoretical Framework

✓ REQUIRED — All Students

- Identify one or more established theories, models, or frameworks that provide the theoretical underpinning for your project.
- Explain the theory clearly and then relate it explicitly to your project — do not just describe the theory in isolation.
- Typical theories/models used in CS projects include: Technology Acceptance Model (TAM), Unified Theory of Acceptance and Use of Technology (UTAUT), Agile development models, OSI Model, CAP Theorem, Actor-Network Theory, Shannon-Weaver Communication Model, etc.
- Typical length: 400 – 700 words.

◈ OPTIONAL — Include Only if Applicable to Your Topic

- If your project is a purely engineering/systems-development project with no social or behavioural dimension, you may frame your 'theoretical framework' as the 'design paradigm' or 'engineering model' (e.g., software development life cycle model, network design principles). Consult your supervisor.

2.4 Empirical Review of Related Works

✓ REQUIRED — All Students

- Review at least 12 – 20 prior research studies directly related to your topic.

- For each study reviewed, cover: the authors and year; the problem addressed; the methodology used; the findings/results; and the relevance to your project.
- Do NOT simply list summaries one after another. Group related studies thematically or chronologically and draw connections between them.
- Critically evaluate each study — what did it do well? What were its shortcomings? How does your project build on or differ from it?
- Typical length: 1,500 – 2,500 words.

✗ COMMON MISTAKES TO AVOID

- Reviewing studies that are not directly related to your topic simply to inflate your reference count.
- Writing 'According to [Author], [Year], the study found that...' for every single paper — vary your sentence construction.
- Copying abstract paragraphs from papers without citation (plagiarism) or without critical engagement.
- Reviewing papers without stating their relevance to your own project.

2.5 Review of Existing Systems / Tools

✓ REQUIRED — All Students

- Identify and critically review two to five (2–5) existing systems, applications, or tools that are similar to what you intend to build.
- For each existing system, evaluate: core functionality; technologies used; known strengths; known weaknesses or gaps; and why it does not fully address your problem.
- This section justifies why a new system is needed.
- Typical length: 600 – 1,000 words.

◈ OPTIONAL — Include Only if Applicable to Your Topic

- If your project is purely a research/experimental study with no system or tool to compare against, this section may be replaced with a 'Review of Related Methodologies and Tools' section covering relevant algorithms, frameworks, or experimental approaches.

2.6 Comparative Analysis of Related Works

✓ REQUIRED — All Students

- Present a comparative table that summarises the key attributes of the studies or systems reviewed in Sections 2.4 and 2.5.
- The table should allow the reader to see at a glance: what each work did, the approach used, the dataset or context, performance metrics, and limitations.
- Follow the table with a paragraph discussing what patterns, trends, and gaps the comparison reveals.
- Typical table: 5–15 rows, 5–7 columns.

✦ WRITING TIP

Label your comparative table appropriately (e.g., 'Table 2.1: Comparative Analysis of Existing Works on [Your Topic]') and reference it in your text before it appears. This table is often what examiners look at first — make it comprehensive and informative.

2.7 Identified Research Gap

✓ REQUIRED — All Students

- Based on your review in Sections 2.4, 2.5, and 2.6, explicitly state the gap(s) in existing knowledge or practice that your project addresses.
- The gap must be specific, defensible, and directly connected to what you will do in Chapters Three and Four.
- Do NOT simply say 'no one has done this before' — explain precisely what aspect has been overlooked, underperformed, or not applied in a particular context.
- Typical length: 200 – 400 words.

2.8 Summary of the Chapter

✓ REQUIRED — All Students

- Synthesise the key insights from the entire chapter.
- Restate the identified gap and explain how your project positions itself within the existing body of work.
- Transition the reader naturally toward Chapter Three (your methodology).
- Typical length: 150 – 250 words.

2.9 Review of Relevant Algorithms [OPTIONAL]

◈ OPTIONAL — Include Only if Applicable to Your Topic

- INCLUDE ONLY FOR: AI, Machine Learning, Data Science, Computer Vision, Natural Language Processing, and Algorithm-intensive projects.
- Review the specific algorithms considered for your project (e.g., Random Forest vs. SVM vs. CNN for image classification).
- For each algorithm: explain how it works, its advantages, its limitations, and why you selected it (or why you did not).
- Include relevant mathematical notation where necessary, with all variables defined.

2.10 Review of Relevant Standards and Protocols [OPTIONAL]

◈ OPTIONAL — Include Only if Applicable to Your Topic

- INCLUDE ONLY FOR: Network engineering, cybersecurity, IoT, communication systems, and protocol-based projects.
- Review the technical standards (IEEE standards, RFC specifications, OWASP guidelines, etc.) relevant to your system.
- Explain how adherence to these standards shapes your design choices in Chapter Three.

2.11 Review of Relevant Datasets [OPTIONAL]

◈ OPTIONAL — Include Only if Applicable to Your Topic

- INCLUDE ONLY FOR: Projects that use existing datasets (machine learning, data analytics, NLP, computer vision).
- Describe each dataset considered: source, size, format, class distribution, collection method, licence.
- Justify why the selected dataset is appropriate for your research objectives.
- Identify any known biases or limitations of the dataset and how you will address them.

Chapter	Section	Min. Words	Max. Words
Chapter Two	Conceptual Review	600	1,000
Chapter Two	Theoretical Framework	400	700
Chapter Two	Empirical Review of Related Works	1,500	2,500
Chapter Two	Review of Existing Systems	600	1,000
Chapter Two	Comparative Analysis + Gap	400	700
Chapter Two	Optional sections (if included)	400	800
Chapter Two	TOTAL (estimated)	3,500	6,700

CHAPTER THREE — METHODOLOGY

Chapter Three is the technical core of your project. It details HOW you conducted your research or developed your system. It must be detailed enough that a competent person in the field could replicate your work. Nothing in this chapter should come as a surprise — everything here must be justified in terms of the objectives stated in Chapter One.

⚠ IMPORTANT NOTICE

Chapter Three must be written before implementation begins (as a plan) and then updated to reflect what was actually done. Inconsistencies between Chapter Three (what you said you would do) and Chapter Four (what you actually did) are a serious red flag for examiners and suggest either poor planning or dishonest reporting.

3.1 Overview of Sub-sections

Sub-Section	Applicable To	Status
3.1 Introduction to the Chapter	All topics	**Required**
3.2 Research Design / Project Approach	All topics	**Required**
3.3 Analysis of Existing System	System-development topics	**Required**
3.4 Proposed System Overview	System-development topics	**Required**
3.5 System Requirements	System-development topics	**Required**
3.6 Data Collection Methods	Research-based topics	**Required**
3.7 Population and Sampling	Survey/empirical studies	**Optional**
3.8 System Architecture / Design	All topics	**Required**
3.9 Use Case / UML Diagrams	System-development topics	**Required**
3.10 Database Design	Data-driven systems	**Required**
3.11 Algorithm / Model Design	AI/ML/Research topics	**Optional**
3.12 Tools and Technologies	All topics	**Required**
3.13 Ethical Considerations	All topics	**Required**
3.14 Summary of the Chapter	All topics	**Required**

3.1 Introduction to the Chapter

✓ REQUIRED — All Students

- Briefly introduce the chapter and what it covers.
- Restate the aim and objectives (from Chapter One) as the basis for the methodology.

- Typical length: 100 – 200 words.

3.2 Research Design / Project Approach

✓ REQUIRED — All Students

- State and justify the overall research design or project approach adopted.
- For system development projects: identify and justify the Software Development Life Cycle (SDLC) model used — Agile/Scrum, Waterfall, Iterative, Rapid Application Development (RAD), etc.
- For research/experimental projects: state whether the study is quantitative, qualitative, or mixed-methods, and justify the choice.
- Explain why the chosen approach is the most appropriate for your specific objectives.
- Typical length: 300 – 500 words.

✦ WRITING TIP

Do not just name a methodology (e.g., 'Agile was used'). Explain what Agile entails, why its features (iterative development, sprint cycles, flexibility) suit your project, and how you specifically applied it in practice (what were your sprints? what did each produce?).

3.3 Analysis of the Existing System

✓ REQUIRED — All Students

- Describe the current state of the system, process, or situation your project seeks to improve.
- Identify specific problems, bottlenecks, inefficiencies, or shortcomings of the existing arrangement.
- Use diagrams where appropriate (e.g., a flowchart of the current manual process).
- This section provides the 'before' picture that your system will improve upon.
- Typical length: 300 – 600 words.

◈ OPTIONAL — Include Only if Applicable to Your Topic

- For projects that address a completely new problem (where no prior system exists), replace this section with 'Problem Domain Analysis' — a detailed technical analysis of the domain in which the problem exists.

3.4 Proposed System Overview

✓ REQUIRED — All Students

- Provide a high-level description of the system or solution you are building.

- Explain what the system does, who uses it, and how it resolves the problems identified in Section 3.3.
- Include a high-level system block diagram or architecture overview.
- Typical length: 300 – 500 words.

3.5 System Requirements

✓ REQUIRED — All Students

- List all Functional Requirements (FR): what the system must DO — specific features and capabilities. Format as: FR1, FR2, FR3, etc.
- List all Non-Functional Requirements (NFR): qualities the system must HAVE — performance, security, usability, scalability, reliability. Format as: NFR1, NFR2, etc.
- Requirements must be specific and testable. 'The system must be fast' is NOT acceptable. 'The system must return search results in under 2 seconds for datasets up to 10,000 records' IS acceptable.
- Hardware and software requirements for deployment must also be listed.

3.6 Data Collection Methods

✓ REQUIRED — All Students

- Describe how data was or will be collected for the project.
- For system development: describe how user requirements were gathered (interviews, questionnaires, observation, document analysis).
- For research/experimental projects: describe the primary data collection instruments (surveys, experiments, API data extraction, sensor data, scraping with consent).
- For projects using existing datasets: refer to Section 2.11 and explain the data pre-processing steps applied.
- Justify each method chosen and acknowledge any associated limitations.

3.7 Population, Sample, and Sampling Technique [OPTIONAL]

◈ OPTIONAL — Include Only if Applicable to Your Topic

- INCLUDE ONLY IF: Your project involves human participants (surveys, usability testing, interviews, controlled experiments with users).
- Define the target population clearly.
- State the sample size and justify it statistically or practically.
- Identify the sampling technique used (simple random, stratified, purposive, convenience) and justify the choice.
- State the inclusion and exclusion criteria for participants.
- Attach the questionnaire/interview guide in the Appendix and reference it here.

3.8 System Architecture

✓ REQUIRED — All Students

- Present a detailed architectural diagram of your system (e.g., a layered architecture, client-server model, microservices diagram, pipeline diagram for ML).
- Label all components clearly and describe the role of each layer or module.
- Describe the data flow between components.
- Justify the architectural choice — why is this architecture appropriate for your requirements?
- All diagrams must be labelled (e.g., 'Figure 3.1: System Architecture Diagram') and referenced in the text.

3.9 UML and System Diagrams

✓ REQUIRED — All Students

- Use Case Diagram: show all system actors and the functions they can perform. REQUIRED for all system-development projects.
- Activity Diagram / Process Flowchart: show the step-by-step flow of key processes. REQUIRED.
- Sequence Diagram: show the interaction between objects/components for key use cases. Required for web/API/multi-tier systems.
- Class Diagram: show system classes, their attributes, methods, and relationships. Required for OOP-based systems.
- Entity-Relationship (ER) Diagram: show the data model. Required if the system uses a relational database (see Section 3.10).
- All diagrams must follow proper UML notation. Use tools such as draw.io, StarUML, Lucidchart, or Visual Paradigm.
- Each diagram must have a title and figure number and must be referenced in the body text.

◈ OPTIONAL — Include Only if Applicable to Your Topic

- State Machine Diagram — for projects with complex state transitions (embedded systems, protocol design).
- Deployment Diagram — for cloud-deployed systems or distributed architectures.
- Component Diagram — for microservices or modular architectures.

3.10 Database Design

✓ REQUIRED — All Students

- Present and explain the Entity-Relationship (ER) Diagram for your database.
- List all entities, their attributes, and the relationships between them (one-to-one, one-to-many, many-to-many).

- Present the relational schema (table definitions with primary keys and foreign keys clearly indicated).
- Justify your choice of database management system (MySQL, PostgreSQL, MongoDB, etc.).

◈ OPTIONAL — Include Only if Applicable to Your Topic

- OMIT this section if your project does not use a database (e.g., a pure algorithm study, a compiler project, or a simulation study). Replace with a 'Data Structure Design' section describing the data structures and memory management approach used.
- For NoSQL (document, graph, key-value) databases: adapt this section to describe the document schema or graph model instead of a relational schema.

3.11 Algorithm / Model Design [OPTIONAL]

◈ OPTIONAL — Include Only if Applicable to Your Topic

- INCLUDE ONLY FOR: AI/ML projects, computer vision, NLP, algorithm-comparison studies, or any project where the core contribution is a new or adapted algorithm.
- Present the algorithm in pseudocode or as a numbered step-by-step procedure.
- Explain the algorithmic complexity (time and space complexity using Big-O notation) where relevant.
- For ML projects: describe the model architecture (layers, activation functions, hyperparameters); the training strategy (loss function, optimiser, learning rate); cross-validation approach; and performance metrics (accuracy, F1-score, RMSE, etc.).
- Justify every significant design decision.

3.12 Tools and Technologies

✓ REQUIRED — All Students

- List and justify every significant tool, language, framework, library, and platform used in the project.
- Present as a structured table: Tool | Version | Purpose | Justification.
- Do not simply list tools — for each major tool, briefly explain WHY it was chosen over alternatives.
- Separate into categories: Programming Language(s); Framework(s); Database(s); Development Environment; Testing Tools; Deployment Platform; Design/Modelling Tools.

3.13 Ethical Considerations

✓ REQUIRED — All Students

- Address the ethical implications of your project, regardless of topic.

- For projects involving human participants: confirm that ethical approval was obtained; that informed consent was secured; that data will be anonymised and stored securely; and that participants had the right to withdraw.
- For AI/ML projects: address potential bias in training data and how it was mitigated; discuss the fairness and explainability of the model.
- For security projects: confirm that penetration testing or vulnerability scanning was conducted only on systems you own or have explicit written permission to test.
- For all projects: address data privacy compliance (Nigeria Data Protection Act 2023), intellectual property rights of third-party code/data used, and any other applicable ethical dimension.
- Typical length: 200 – 400 words.

3.14 Summary of the Chapter

✓ REQUIRED — All Students

- Summarise the key design and methodological decisions made in this chapter.
- Briefly restate what will be implemented and tested in Chapter Four.
- Typical length: 100 – 200 words.

Chapter	Section	Min. Words	Max. Words
Chapter Three	Research Design	300	500
Chapter Three	Analysis & Proposed System	600	1,100
Chapter Three	System Requirements	400	700
Chapter Three	Data Collection	300	500
Chapter Three	Architecture & Diagrams	600	1,000
Chapter Three	DB / Algorithm Design	400	800
Chapter Three	Tools & Ethics	400	700
Chapter Three	TOTAL (estimated)	3,000	5,300

CHAPTER FOUR — IMPLEMENTATION

Chapter Four demonstrates that the system or solution designed in Chapter Three has been fully built, validated, and evaluated. It is the proof-of-concept chapter — here you show that the system works, that it meets the specified requirements, and that its performance has been rigorously evaluated. Results must be honest; do not report only success cases.

⚠ IMPORTANT NOTICE

A working system is MANDATORY. A system that crashes, produces incorrect outputs, or cannot be demonstrated during the project defence will attract significant mark deductions (up to 20 marks from the implementation score). Ensure your system is fully functional and tested before submission. All screenshots must be of your actual running system — not mockups or design wireframes.

4.1 Overview of Sub-sections

Sub-Section	Applicable To	Status
4.1 Introduction to the Chapter	All topics	**Required**
4.2 Implementation Environment	All topics	**Required**
4.3 System Implementation	System-development topics	**Required**
4.4 System Interface / Screenshots	System-development topics	**Required**
4.5 Test Plan	All topics	**Required**
4.6 Test Cases and Test Results	All topics	**Required**
4.7 Model Training and Evaluation	AI/ML/Data Science topics	**Optional**
4.8 Experimental Results and Analysis	Research/Algorithm topics	**Optional**
4.9 Performance Evaluation	All topics	**Required**
4.10 Discussion of Results	All topics	**Required**
4.11 Summary of the Chapter	All topics	**Required**

4.1 Introduction to the Chapter

✓ REQUIRED — All Students

- Briefly introduce the chapter, linking it to the methodology described in Chapter Three.
- Typical length: 80 – 150 words.

4.2 Implementation Environment

✓ REQUIRED — All Students

- Describe the hardware and software environment in which the system was developed and tested.
- Include: processor, RAM, storage, operating system, development tools, and deployment platform.
- Present as a structured table for clarity.

4.3 System Implementation

✓ REQUIRED — All Students

- Describe the implementation of each major module or component of your system.
- For each module: state its purpose; describe the key implementation decisions; present the most important code snippet (core logic only — not trivial boilerplate).
- Code snippets in the document must be in Courier New, 10pt, single-spaced, inside a clearly bordered box.
- Complete source code goes in the Appendix and on the digital submission — NOT all in the main document.
- Reference all code snippets properly (e.g., 'Listing 4.1: User Authentication Function').
- Typical length (prose + snippets combined): 1,000 – 2,000 words.

✦ WRITING TIP

Focus on the 'interesting' parts of the implementation — the sections that were challenging, that required non-trivial decisions, or that represent your core contribution. There is no value in reproducing hundreds of lines of standard CRUD code. Show the examiner the intelligent parts.

4.4 System Interfaces and Screenshots

✓ REQUIRED — All Students

- Present annotated screenshots of every major interface (screen/page/view) of your system.
- Screenshots must show the system in operation — not placeholder or dummy content.
- Each screenshot must have a figure number and descriptive caption (e.g., 'Figure 4.3: User Registration Interface').
- Annotate screenshots where necessary to draw attention to key features.
- Organise screenshots logically — follow the user journey from login to core features.
- Minimum number of screenshots: 8. Maximum recommended: 20 (additional screens may go in the Appendix).

4.5 Test Plan

✓ REQUIRED — All Students

- Define your overall testing strategy: what types of testing were conducted (unit, integration, system, user acceptance, performance, security)?
- State the testing objectives — what properties of the system were being verified?
- Identify who performed the testing (developer self-testing, peer testing, end-user testing).
- Describe the testing environment (same as or different from the development environment?).

4.6 Test Cases and Test Results

✓ REQUIRED — All Students

- Present a minimum of fifteen (15) structured test cases covering all functional requirements.
- Each test case must include: Test Case ID; Test Objective; Preconditions; Test Input(s); Expected Output; Actual Output; Pass/Fail status.
- Present test cases in a table (e.g., 'Table 4.1: System Test Cases and Results').
- Test cases must cover normal scenarios, boundary cases, and error/invalid input scenarios.
- If any test case FAILED, document it honestly and explain what was done to resolve it (or note it as a limitation if unresolved).

✗ COMMON MISTAKES TO AVOID

- Reporting 100% pass rate on all test cases without any discussion — examiners will question whether testing was rigorous.
- Writing test cases that are too vague to be reproducible (e.g., 'Test: Click login button. Result: Works').
- Omitting error-handling and boundary tests.

4.7 Model Training and Evaluation [OPTIONAL]

◈ OPTIONAL — Include Only if Applicable to Your Topic

- INCLUDE ONLY FOR: Machine Learning, Deep Learning, NLP, and Computer Vision projects.
- Report the training configuration: dataset split (train/validation/test ratios), number of epochs, batch size, optimiser, learning rate schedule.
- Present training and validation loss/accuracy curves (as figures).
- Report final model performance on the held-out test set using appropriate metrics: Accuracy, Precision, Recall, F1-Score, AUC-ROC, RMSE, BLEU score, etc.
- Present a confusion matrix for classification problems.
- Compare your model's performance with at least two (2) baseline models or prior works from your literature review.
- Analyse misclassifications or errors — what types of cases does the model struggle with?

4.8 Experimental Results and Analysis [OPTIONAL]

◈ OPTIONAL — Include Only if Applicable to Your Topic

- INCLUDE ONLY FOR: Algorithm comparison studies, simulation experiments, network performance studies, or any project where the contribution is primarily empirical.
- Present results in tables and graphs — use bar charts, line graphs, scatter plots, or box plots as appropriate.
- Ensure all axes are labelled, units are stated, and figures are numbered and captioned.
- Report statistical measures where applicable: mean, standard deviation, confidence intervals.
- If hypotheses were stated in Section 1.6, state clearly here whether each is accepted or rejected, based on the data.

4.9 Performance Evaluation

✓ REQUIRED — All Students

- Evaluate the system against the Non-Functional Requirements specified in Section 3.5.
- Test and report on: response time / processing speed; system accuracy or correctness; resource utilisation (CPU, memory, storage); scalability (if testable); security (basic vulnerability checks); usability (if a user study was conducted).
- Present results quantitatively where possible — use tables and charts.
- If a user satisfaction survey was conducted: present the questionnaire results (e.g., Likert scale ratings) in charts and interpret them.

4.10 Discussion of Results

✓ REQUIRED — All Students

- Interpret your results — do not simply restate what the tables show. Explain WHAT the results mean.
- Relate findings back to your research questions and objectives from Chapter One.
- Compare your results with findings from related works reviewed in Chapter Two — are your results consistent, better, or worse? Why?
- Acknowledge unexpected results and attempt to explain them.
- Typical length: 400 – 700 words.

✦ WRITING TIP

The Discussion section is where your intellectual contribution becomes most visible. Moving from data to interpretation to meaning is what separates a good project from an excellent one. Ask yourself: 'So what?' after every finding, and make sure your text answers that question.

4.11 Summary of the Chapter

✓ REQUIRED — All Students

- Briefly summarise what was implemented and what the test results demonstrated.
- State whether the system meets its specified requirements.
- Transition to Chapter Five.
- Typical length: 100 – 200 words.

Chapter	Section	Min. Words	Max. Words
Chapter Four	Implementation Environment	150	300
Chapter Four	System Implementation (prose)	1,000	2,000
Chapter Four	Test Plan & Test Cases	500	900
Chapter Four	ML Training/Experimental Results	600	1,200
Chapter Four	Performance Evaluation	400	700
Chapter Four	Discussion of Results	400	700
Chapter Four	TOTAL (estimated)	3,050	5,800

CHAPTER FIVE — SUMMARY, CONCLUSION AND RECOMMENDATIONS

Chapter Five brings the project to a close. It synthesises what was accomplished, reflects on the journey, draws lessons, and opens doors for future work. It must be written with the same care and scholarly rigour as the preceding chapters — it is not a formality; it is your final academic statement.

⚠ IMPORTANT NOTICE

Chapter Five must NOT introduce new findings, new data, or new arguments that have not already been discussed in the preceding chapters. Its purpose is synthesis and reflection, not new content. Every objective stated in Chapter One must be explicitly addressed here.

5.1 Overview of Sub-sections

Sub-Section	Applicable To	Status
5.1 Introduction to the Chapter	All topics	**Required**
5.2 Summary of the Study	All topics	**Required**
5.3 Achievement of Objectives	All topics	**Required**
5.4 Contributions to Knowledge/Practice	All topics	**Required**
5.5 Conclusion	All topics	**Required**
5.6 Limitations of the Study	All topics	**Required**
5.7 Recommendations	All topics	**Required**
5.8 Suggestions for Future Work	All topics	**Required**
5.9 AI Use Disclosure	If AI tools were used	**Optional**

5.1 Introduction to the Chapter

✓ REQUIRED — All Students

- Briefly state the purpose of the chapter and its structure.
- Typical length: 60 – 100 words.

5.2 Summary of the Study

✓ REQUIRED — All Students

- Provide a concise, chapter-by-chapter summary of the entire project.
- For each chapter, describe in 2–4 sentences what was covered and what was found or produced.
- Do NOT copy paragraphs from previous chapters — write fresh, synthesised summaries.
- Typical length: 400 – 600 words.

✗ COMMON MISTAKES TO AVOID

- Simply copying the introductory paragraphs of each chapter verbatim — this is lazy and will be penalised.
- Summarising only the technical chapters and skipping the literature review or introduction.

5.3 Achievement of Objectives

✓ REQUIRED — All Students

- Revisit each objective stated in Section 1.4 and explicitly state whether it was FULLY ACHIEVED, PARTIALLY ACHIEVED, or NOT ACHIEVED.
- For each objective, provide evidence from Chapters Three and Four that supports your claim.
- If any objective was not fully achieved, explain the reason honestly and discuss its impact on the project.
- Present this section as a structured table followed by brief prose discussion.

✦ WRITING TIP

Structure: 'Objective 1 was to [state objective]. This was achieved by [what you did in Chapter 3/4]. Evidence of achievement is demonstrated in [Figure/Table X].' Repeat for each objective.

5.4 Contributions to Knowledge / Practice

✓ REQUIRED — All Students

- Articulate what is NEW about your project — what does it add to the existing body of knowledge or professional practice that was not there before?
- Be specific. Examples: 'This project is the first to apply [technique] to [problem] in the Nigerian context'; 'The proposed algorithm outperforms the benchmark by X% on [dataset]'; 'The developed system reduces processing time from [X] to [Y].'
- Contributions may be theoretical, practical/applied, methodological, or social.
- Typical length: 200 – 400 words.

5.5 Conclusion

✓ REQUIRED — All Students

- Draw a clear, evidence-based conclusion about what your project has demonstrated.
- Connect the conclusion directly to the stated aim (Section 1.3) — was the aim achieved?
- The conclusion must be grounded in the results and discussion of Chapter Four — no new claims.
- Write with confidence but academic restraint — do not overclaim. Distinguish between what your project demonstrates conclusively and what it merely suggests.
- Typical length: 250 – 400 words.

5.6 Limitations of the Study

✓ REQUIRED — All Students

- Revisit and expand on the limitations first mentioned in Section 1.9, now in light of what actually transpired during the project.
- Discuss limitations that only became apparent during implementation or evaluation.
- For each limitation, state: what it is; how it affected the project; and what would need to change to overcome it in future work.
- Typical length: 200 – 350 words.

Note: *Acknowledging limitations is a mark of scholarly maturity. Examiners expect them. A project that claims no limitations is not credible.*

5.7 Recommendations

✓ REQUIRED — All Students

- Provide concrete, actionable recommendations based on your findings.
- Recommendations should be directed at specific stakeholders: system users, developers, organisations, policymakers, or regulatory bodies.
- Each recommendation must be directly derived from a finding or conclusion in the project — do not make generic recommendations unrelated to your work.
- Minimum of four (4) substantive recommendations.
- Typical length: 250 – 400 words.

5.8 Suggestions for Future Work

✓ REQUIRED — All Students

- Identify at least three (3) directions in which this project can be extended or improved.
- Be specific: 'Future work could incorporate transfer learning to improve the model's performance on low-resource Nigerian languages' is better than 'Future work could improve the system.'
- Suggestions should address identified limitations, unexplored objectives, or natural extensions of the current work.

- Typical length: 200 – 350 words.

5.9 AI Use Disclosure [REQUIRED IF AI TOOLS WERE USED]

◈ OPTIONAL — Include Only if Applicable to Your Topic

- REQUIRED IF: You used any AI-assisted tool (ChatGPT, GitHub Copilot, Grammarly AI, etc.) in any part of the project development or writing process.
- State explicitly: which tool(s) were used; for what specific purpose(s); in which sections or modules; and the extent of use.
- Confirm that all AI-assisted content was reviewed, verified, and edited by you, and that you take full responsibility for the final output.
- Format as a brief, honest statement — typically 100 – 200 words.
- This disclosure is MANDATORY if applicable. Failure to disclose AI use when it occurred constitutes academic dishonesty.

Chapter	Section	Min. Words	Max. Words
Chapter Five	Summary of the Study	400	600
Chapter Five	Achievement of Objectives	300	500
Chapter Five	Contributions	200	400
Chapter Five	Conclusion	250	400
Chapter Five	Limitations & Recommendations	450	750
Chapter Five	Future Work	200	350
Chapter Five	TOTAL (estimated)	1,800	3,000

OVERALL WORD COUNT AND PAGE ESTIMATE

The following table provides the total word count guidance for a complete project report. Note that these are estimates and guides — the quality of content is more important than hitting a specific word count. Examiners will penalise padded, repetitive, or off-topic content more heavily than a report that is slightly shorter but intellectually rigorous.

Chapter	Min. Words	Max. Words	Est. Pages
Chapter One — Introduction	1,950	3,400	8–14
Chapter Two — Literature Review	3,500	6,700	15–27
Chapter Three — Methodology & Design	3,000	5,300	13–22
Chapter Four — Implementation & Testing	3,050	5,800	13–24
Chapter Five — Summary & Conclusion	1,800	3,000	8–12
Front Matter (Abstract, ToC, etc.)	400	700	5–8
References	—	—	4–8
Appendices	—	—	Variable
GRAND TOTAL	**13,700**	**24,900**	**66–115**

Note: *Page estimates assume Times New Roman 12pt, 1.5 line spacing, and standard margins. Figures, tables, and code listings add to the page count but not the word count.*

PRE-SUBMISSION CHAPTER CHECKLIST

Before submitting your project for assessment, work through this checklist for each chapter. Your supervisor should also review this checklist during the final review meeting.

CHAPTER ONE CHECKLIST

☐	Background clearly establishes context and relevance (funnel structure used)
☐	Problem statement is specific, evidenced, and flows from the background
☐	Aim is stated as a single overarching goal with an action verb
☐	Objectives are SMART and numbered (4–7 objectives)
☐	Research questions are directly derived from objectives
☐	Significance addresses at least three stakeholder groups with specific benefits
☐	Scope defines what IS and what IS NOT covered
☐	Limitations are acknowledged with mitigation strategies
☐	All key terms are defined operationally with citations
☐	Organisation of report is clear and accurate

CHAPTER TWO CHECKLIST

☐	Minimum 15 references cited; 70%+ from last 7 years; peer-reviewed sources only
☐	Conceptual review goes beyond definitions — concepts are related to each other and to the project
☐	Theoretical framework identifies a named theory and explicitly links it to this project
☐	Empirical review covers at least 12 related studies, grouped thematically
☐	Each reviewed study is critically evaluated (not just summarised)
☐	Existing systems/tools are reviewed and their shortcomings identified
☐	Comparative analysis table is present, labelled, and discussed
☐	Research gap is explicitly stated and is specific and defensible
☐	Chapter ends with a clear summary and transition to Chapter Three

CHAPTER THREE CHECKLIST

☐	Research design / SDLC model is named and justified (not just described)
☐	Analysis of existing system is documented with diagrams
☐	Proposed system overview includes a high-level architecture diagram
☐	Functional requirements are listed as FR1, FR2… (specific and testable)
☐	Non-functional requirements are listed as NFR1, NFR2… (specific and measurable)
☐	Data collection methods are described and justified
☐	Use case diagram covers all actors and all major system functions
☐	At least one activity/process flowchart is included
☐	ER diagram and relational schema are included (if database is used)
☐	All tools, frameworks, and technologies are listed with justification
☐	Ethical considerations are addressed in a dedicated sub-section
☐	All diagrams are labelled with figure numbers and referenced in the text

CHAPTER FOUR CHECKLIST

☐	Implementation environment (hardware + software) is documented in a table
☐	Key modules are described with code snippets (Courier New 10pt in bordered boxes)
☐	Screenshots cover all major interfaces — minimum 8, all showing real system data
☐	All screenshots have figure numbers and descriptive captions
☐	Test plan defines the testing strategy and types of tests conducted
☐	At least 15 structured test cases are presented in a table
☐	Test cases cover normal, boundary, and error scenarios
☐	Test results are honest — any failures are documented and explained
☐	Performance is evaluated against Non-Functional Requirements
☐	Discussion interprets results and compares with related works

CHAPTER FIVE CHECKLIST

☐	Chapter-by-chapter summary is written freshly (not copy-pasted from earlier chapters)
☐	Each objective from Section 1.4 is explicitly addressed (fully / partially / not achieved)
☐	Contributions to knowledge are specific and defensible

☐	Conclusion is grounded in results and directly addresses the aim
☐	Limitations are honest and include new ones discovered during the project
☐	Recommendations are specific and directed at named stakeholders
☐	At least 3 specific suggestions for future work are provided
☐	AI use is disclosed if any AI tools were used (Section 5.9)
☐	No new findings or arguments are introduced in this chapter

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This guide is issued by the Departmental Project Committee and is subject to annual review.

For queries, contact the Project Coordinator through the Department Secretary.